



INDIVIDUAL ASSIGNMENT

COMP40004

WEB DEVELOPING AND OPERATING SYSTEM - 2 COM24A2

Name - Ganidu Sasmitha Rajapaksha Udage

ID - CB 014572

Web site link – <https://ganidusasmitha.github.io/gamig4world/consoles.html>

Git hub file link - <https://github.com/GaniduSasmitha/gamig4world.git>

Table of Contents

1. Introduction to my web page	3
2. Activity diagrams	3
3. HTML and CSS Validity	5
3.1 Console page.....	5
3.2 Favorite page	6
3.3 Cart page.....	6
3.4 Product page	7
3.5 Payment page.....	8
4. Accessibility using WAVE.....	9
5. Lighthouse testing.....	11
5.1 Desktop view tests	11
5.2 Mobile view tests.....	13
6. Responsiveness on different devices and browsers.....	15
6.1 Mobile view	15
6.2 Tab view	16
6.3 Edge browser view	17
7. Functional tests for your product reservation application	18
8. Conclusion	19

Table of Figures

Figure 1.....	3
Figure 2.....	4
Figure 3.....	4
Figure 4.....	4
Figure 5.....	4
Figure 6.....	5
Figure 7.....	5
Figure 8.....	6
Figure 9.....	6
Figure 10.....	6
Figure 11.....	7
Figure 12.....	7
Figure 13.....	7
Figure 14.....	8
Figure 15.....	8
Figure 16.....	9
Figure 17.....	9
Figure 18.....	10
Figure 19.....	10
Figure 20.....	10
Figure 21.....	11
Figure 22.....	11
Figure 23.....	12
Figure 24.....	12
Figure 25.....	12
Figure 26.....	13
Figure 27.....	13
Figure 28.....	13
Figure 29.....	14
Figure 30.....	14
Figure 31.....	15
Figure 32.....	15
Figure 33.....	15
Figure 34.....	15
Figure 35.....	16
Figure 36.....	16
Figure 37.....	16
Figure 38.....	16
Figure 39.....	17
Figure 40.....	17
Figure 41.....	17

1. Introduction to my web page

This web application is an internet site for the sale of different gaming products. The site is simple in design where clients can browse and purchase gaming consoles, accessories and other hardware. The site provides key features such as adding products to a shopping cart, adding to favorites, and added online payment. The vision of this site is to provide gamers with a simple and reliable shopping experience, whether they are casual users or gamers.

2. Activity diagrams

An activity diagram is a Unified Modeling Language diagram. It shows how activities in a flowchart within a process or system. It provides a graphical representation of the order of steps, decisions, and workflows. This allows one to easily see how a specific function or feature works. Activity diagrams for web applications can often be utilized to model how the users will interact, for example, browsing products, placing items in a cart, and making a purchase. Activity diagrams help in planning, analyzing, and refining the system for an end user. From below figures shows process of my console page buttons from activity diagrams.

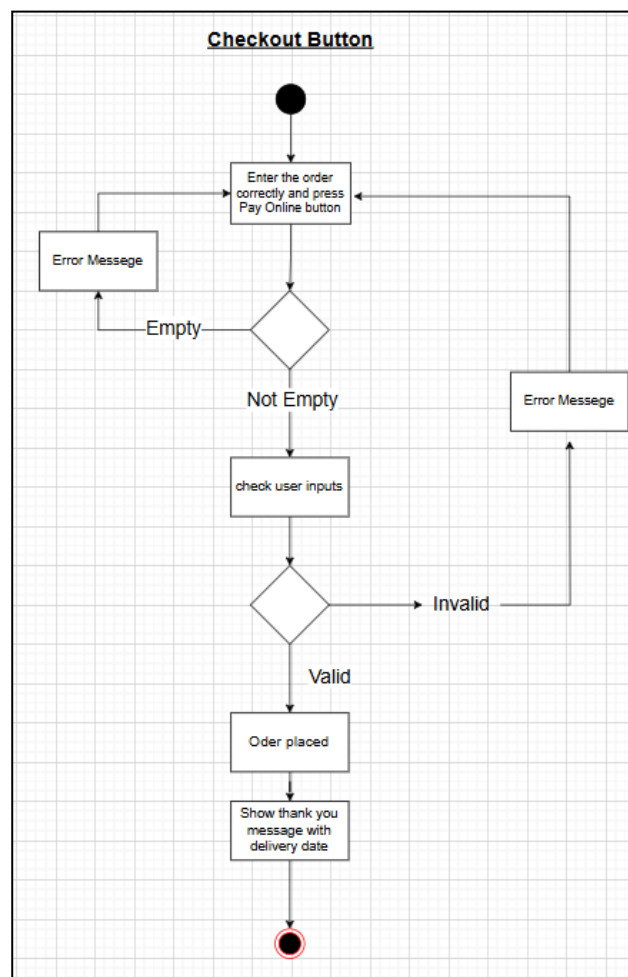


Figure 1

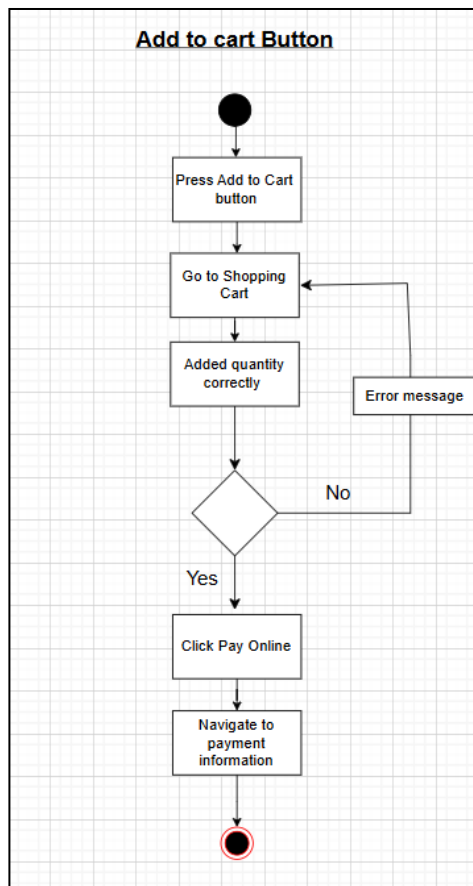


Figure 2

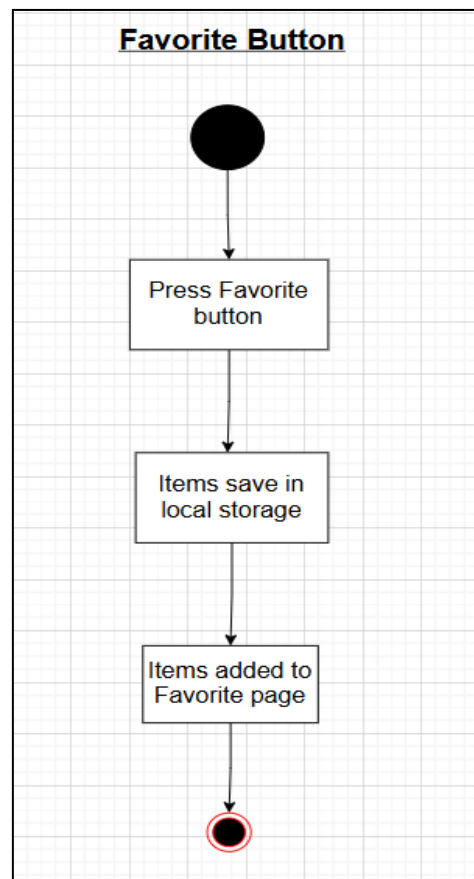


Figure 3

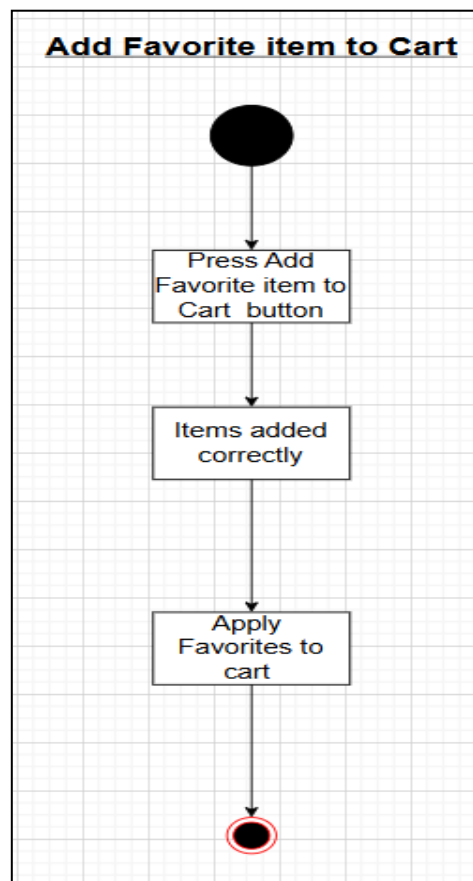


Figure 4

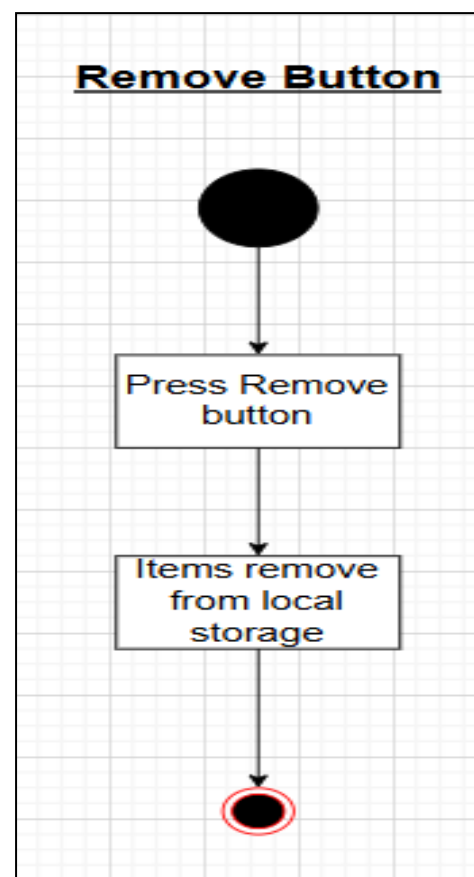


Figure 5

3. HTML and CSS Validity

HTML and CSS validity means that the code utilized to construct and develop a site complies with the official guidelines established by the World Wide Web Consortium. Valid HTML is such that it guarantees the right structure of the web page so that browsers load the content accordingly. Likewise, valid CSS is such that it guarantees the style rules are correctly written so that the design loads exactly the same in different browsers and devices. Checking both HTML and CSS helps to catch coding mistakes like open tags, wrong properties, or bad syntax. The W3C Markup Validation Service and the W3C CSS Validator are generally used to run these checks while designing websites. From below figures show my web HTML and CSS validity reports.

3.1 Console page

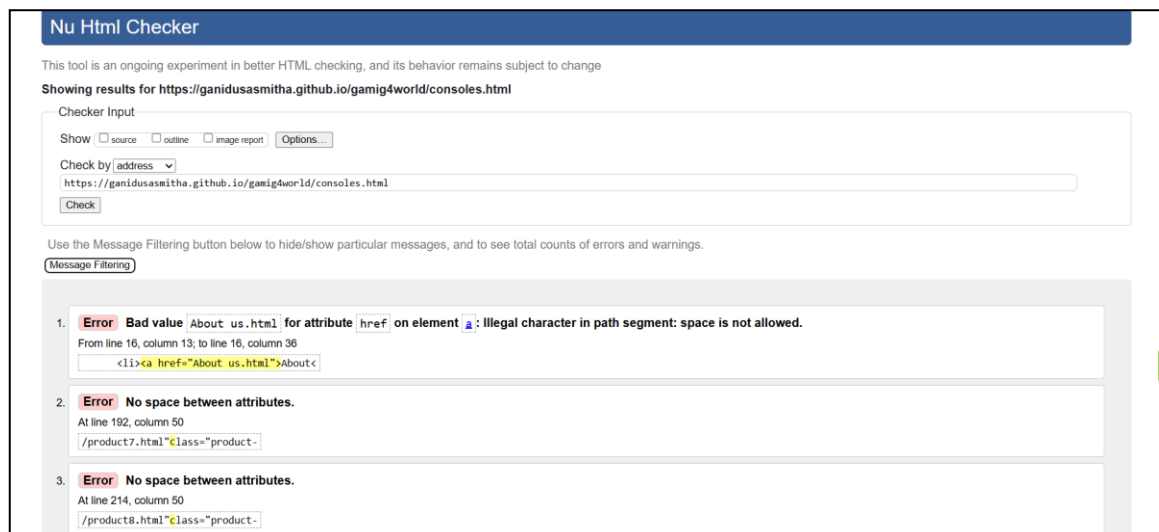


Figure 6

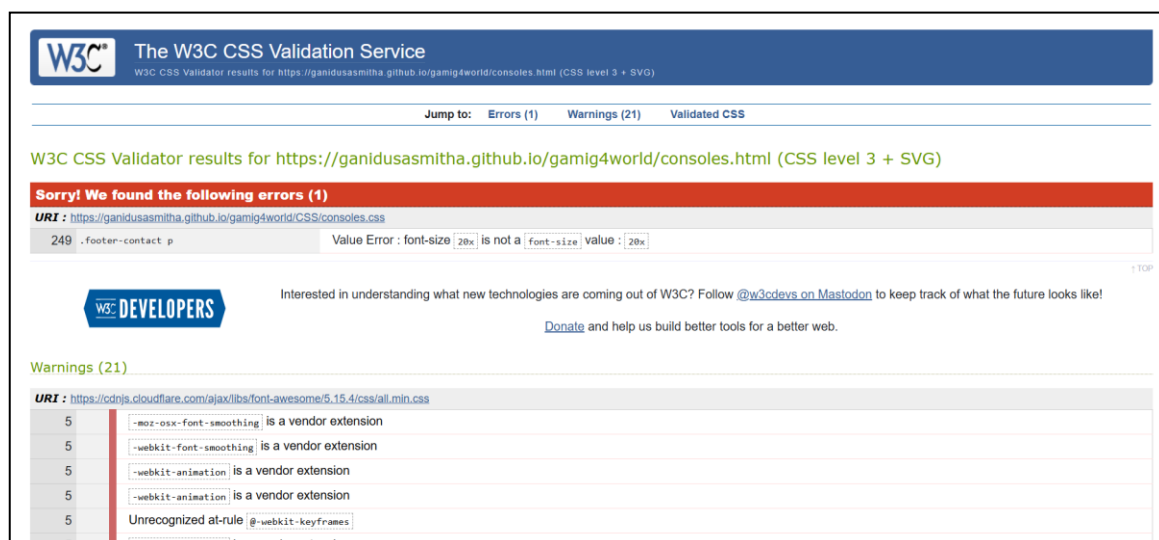
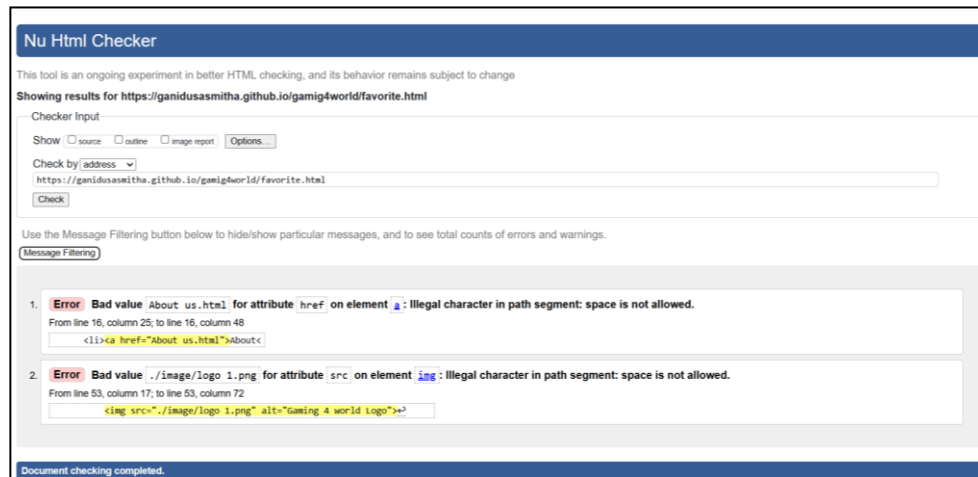


Figure 7

3.2 Favorite page



Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for <https://ganidusasmitha.github.io/gamig4world/favorite.html>

Checker Input

Show ☐ source ☐ outline ☐ image report

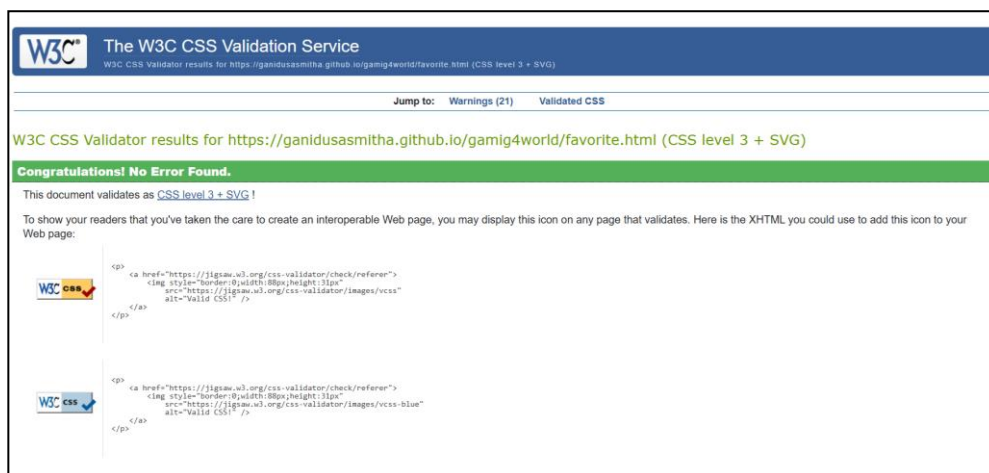
Check by

Use the Message Filtering button below to hide/show particular messages, and to see total counts of errors and warnings.

- Error** Bad value 'About us.html' for attribute 'href' on element 'a': Illegal character in path segment: space is not allowed.
From line 16, column 25; to line 16, column 48
`<li>a href="About us.html">About</li>`
- Error** Bad value './image/logo 1.png' for attribute 'src' on element 'img': Illegal character in path segment: space is not allowed.
From line 53, column 17; to line 53, column 72
``

Document checking completed.

Figure 8



W3C The W3C CSS Validation Service

W3C CSS Validator results for <https://ganidusasmitha.github.io/gamig4world/favorite.html> (CSS level 3 + SVG)

[Jump to:](#) [Warnings \(21\)](#) [Validated CSS](#)

W3C CSS Validator results for <https://ganidusasmitha.github.io/gamig4world/favorite.html> (CSS level 3 + SVG)

Congratulations! No Error Found.

This document validates as **CSS level 3 + SVG** !

To show your readers that you've taken the care to create an interoperable Web page, you may display this icon on any page that validates. Here is the XHTML you could use to add this icon to your Web page:

 `<p>

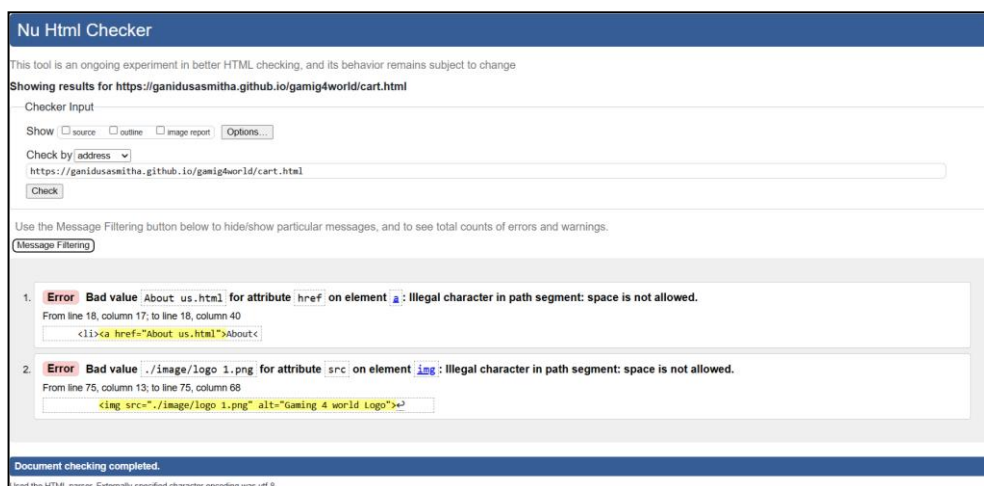
</p>`

 `<p>

</p>`

Figure 9

3.3 Cart page



Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for <https://ganidusasmitha.github.io/gamig4world/cart.html>

Checker Input

Show ☐ source ☐ outline ☐ image report

Check by

Use the Message Filtering button below to hide/show particular messages, and to see total counts of errors and warnings.

- Error** Bad value 'About us.html' for attribute 'href' on element 'a': Illegal character in path segment: space is not allowed.
From line 18, column 17; to line 18, column 40
`<li>a href="About us.html">About</li>`
- Error** Bad value './image/logo 1.png' for attribute 'src' on element 'img': Illegal character in path segment: space is not allowed.
From line 75, column 13; to line 75, column 68
``

Document checking completed.

Used the HTML parser. Externally specified character encoding was utf-8.

Figure 10

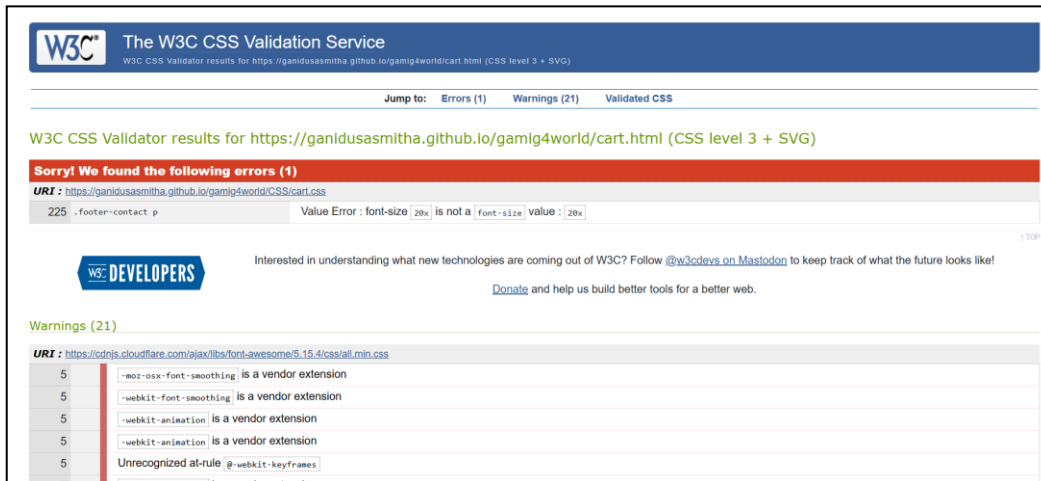


Figure 11

3.4 Product page



Figure 12

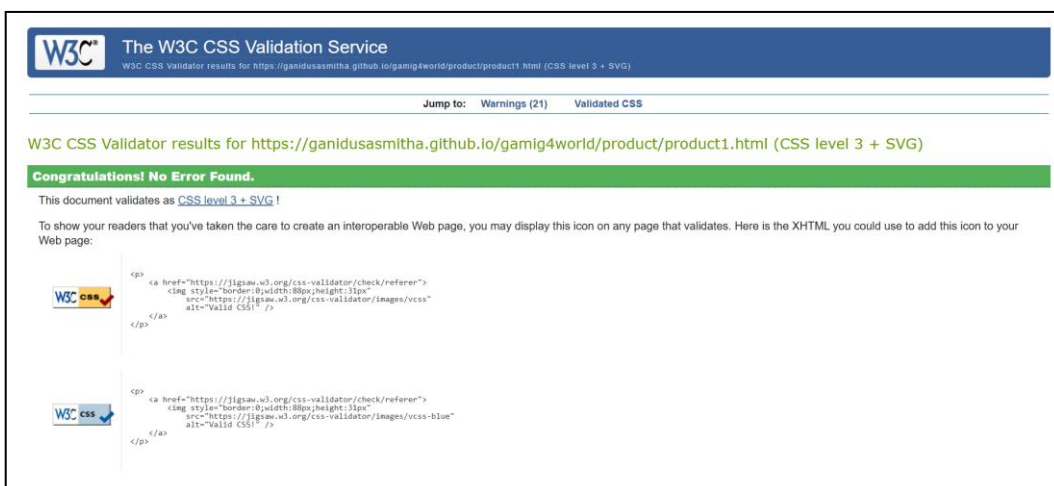


Figure 13

3.5 Payment page

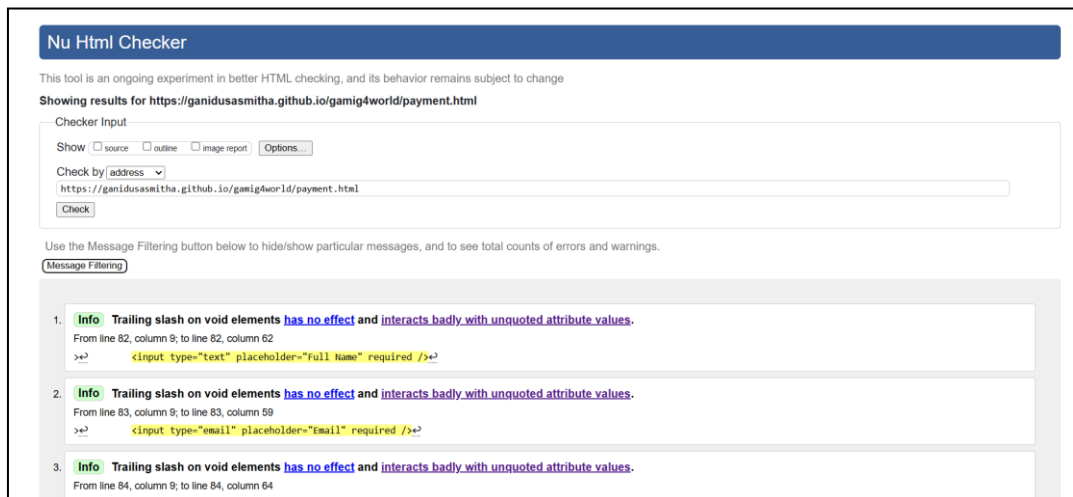


Figure 14

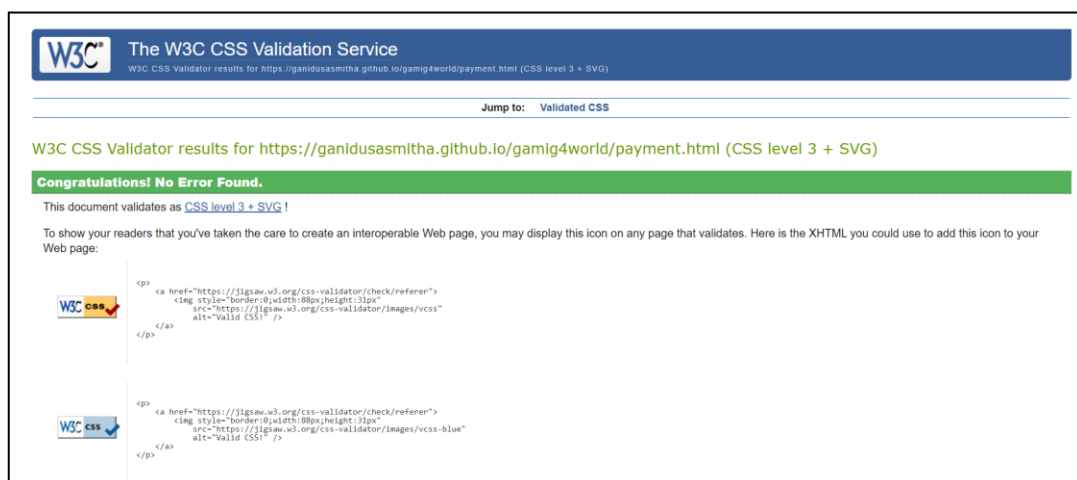


Figure 15

When I was added to my web page HTML validity and CSS validity they show me some errors in my HTML and CSS, you can see that errors from above images. I search that errors and some are fixed.

4. Accessibility using WAVE

WAVE (Web Accessibility Evaluation Tool) is a web tool boiled by Web AIM that helps developers evaluate the accessibility of their websites. It identifies issues that may prevent users with disabilities from accessing or participating in the content. WAVE checks for missing alt text, low contrast between text and background, unlabeled form controls, and improper heading structure. Using WAVE allows developers to assure their websites with accessibility guidelines, ensuring it is inclusive and better for every user. From below figures show my page accessibility wave reports.

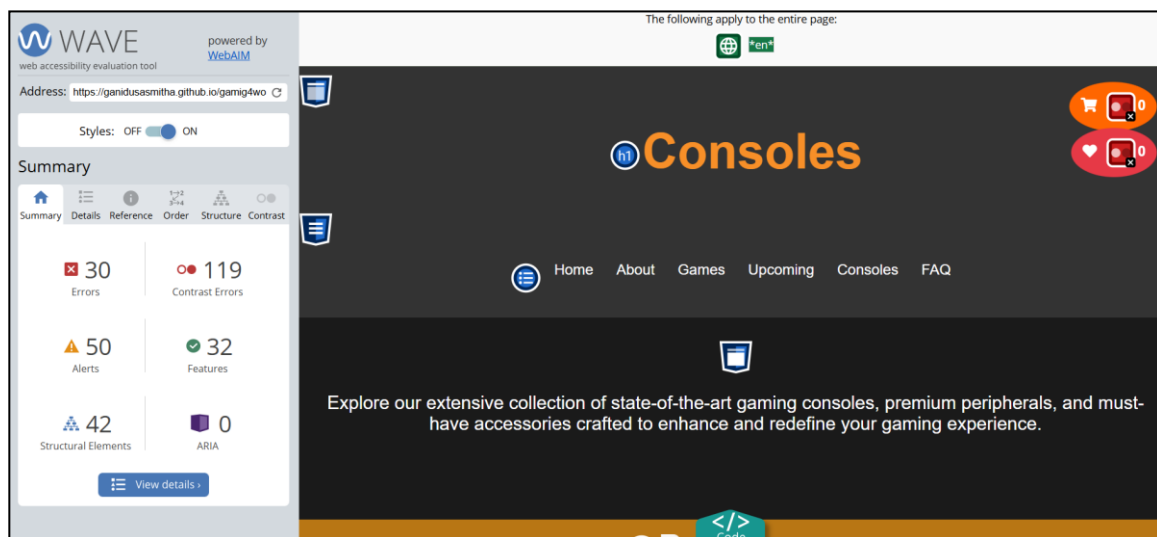


Figure 16

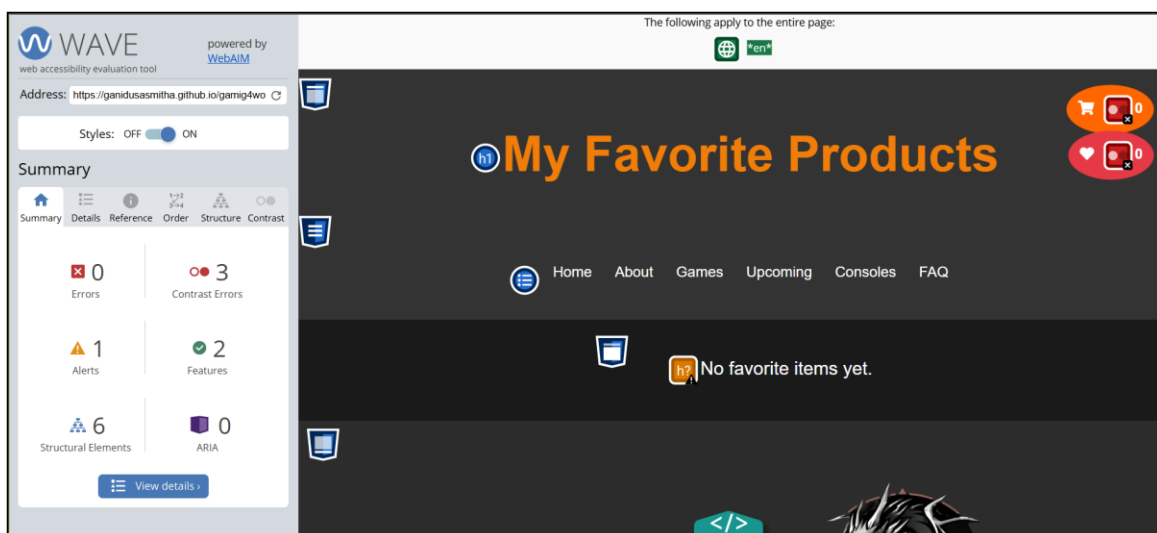


Figure 17

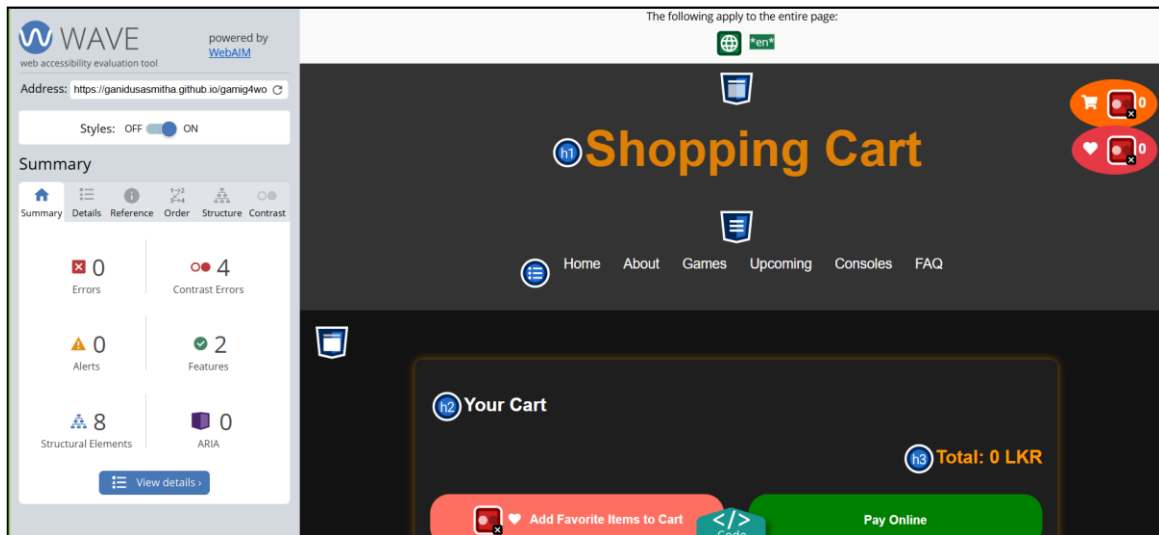


Figure 18

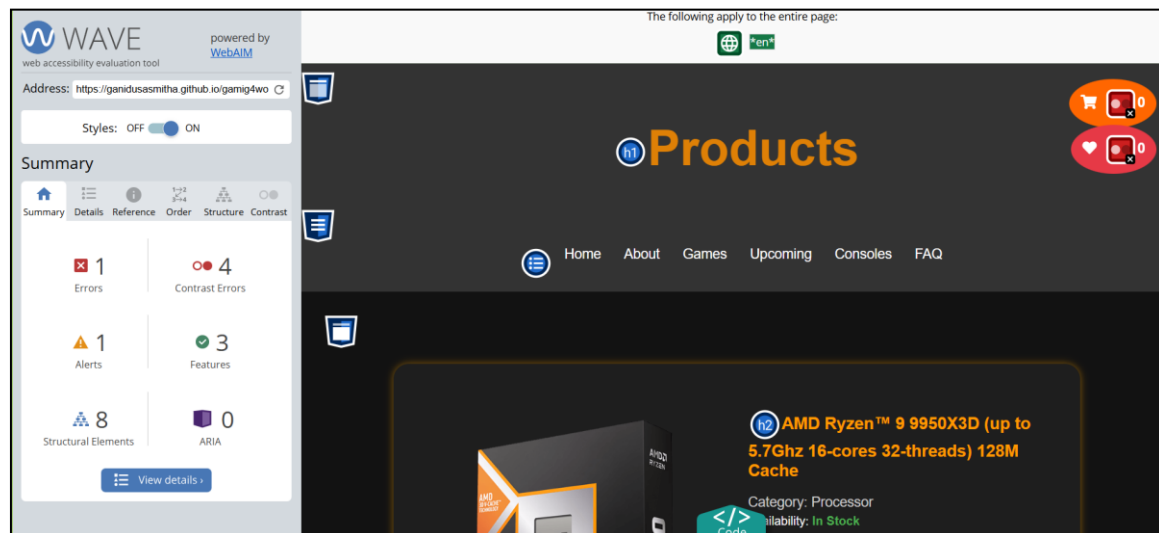


Figure 19

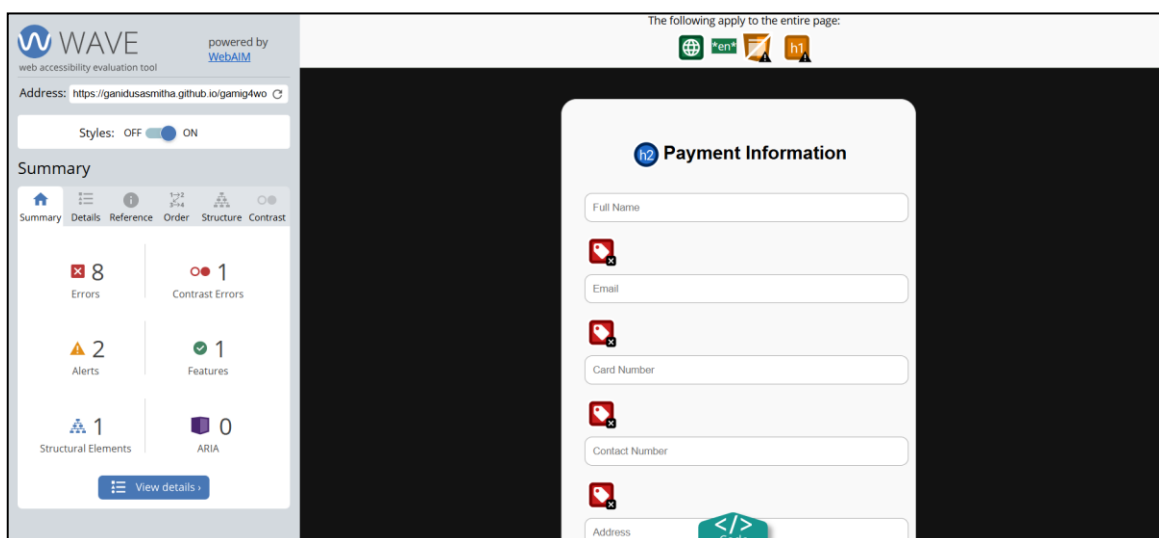


Figure 20

5. Lighthouse testing

Lighthouse is a free, automated Google tool for improving web page quality. It audits a website in certain key areas like performance, accessibility, best practices, SEO (Search Engine Optimization), and Progressive Web App (PWA) features. Lighthouse generates a detailed report with scores and also suggestions on how to do better in each area when run. This helps developers identify issues such as slow loading speeds, accessibility problems, and insufficient metadata. Lighthouse testing is commonly used through Chrome Dev Tools and is helpful in optimizing user experience along with technical quality of a website. From below figures show my website lighthouse tests in mobile and desktop view.

5.1 Desktop view tests

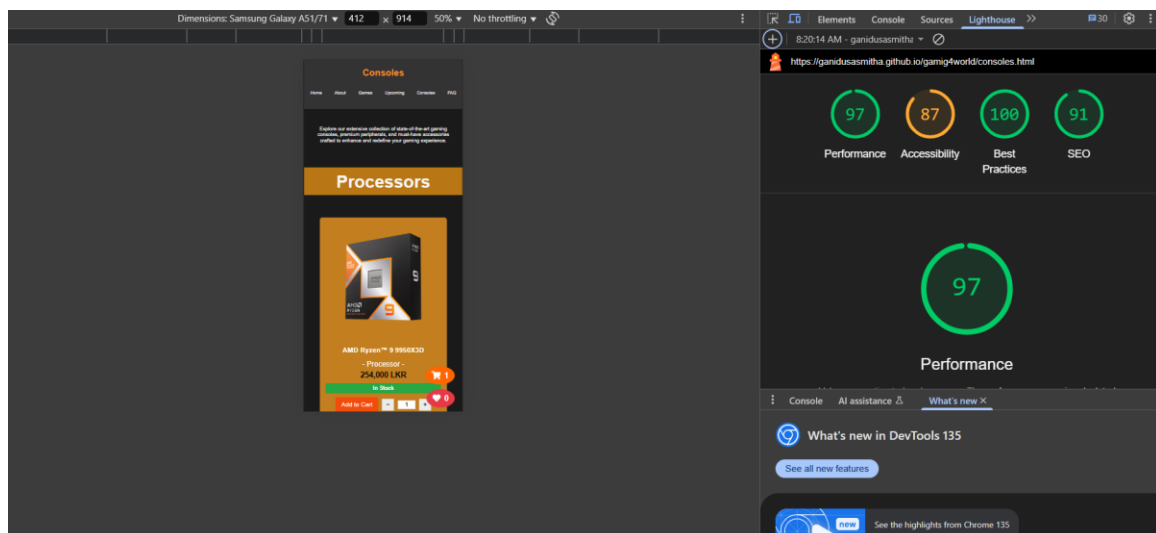


Figure 21

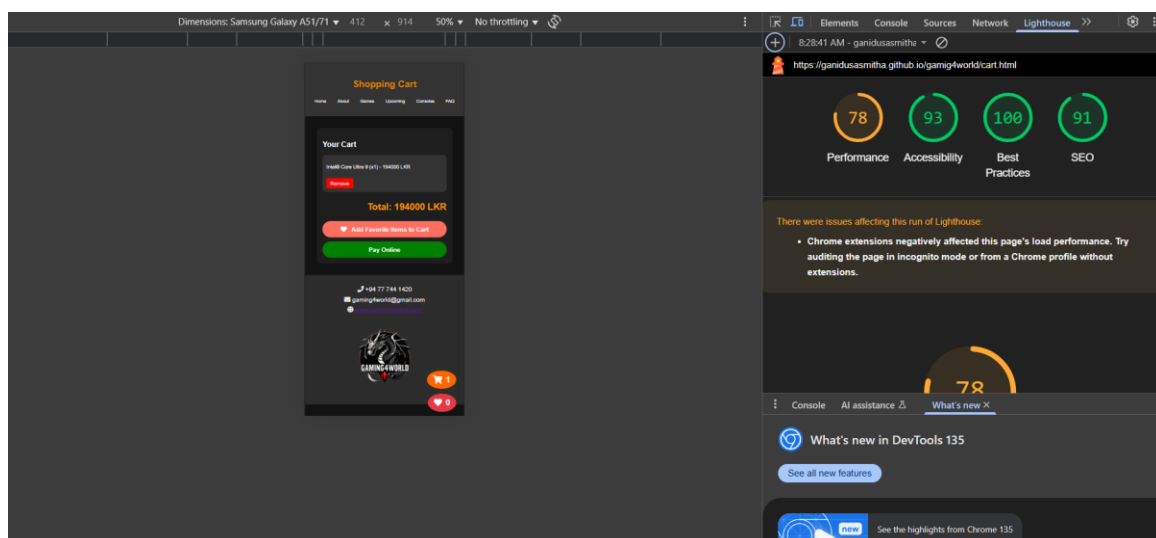


Figure 22

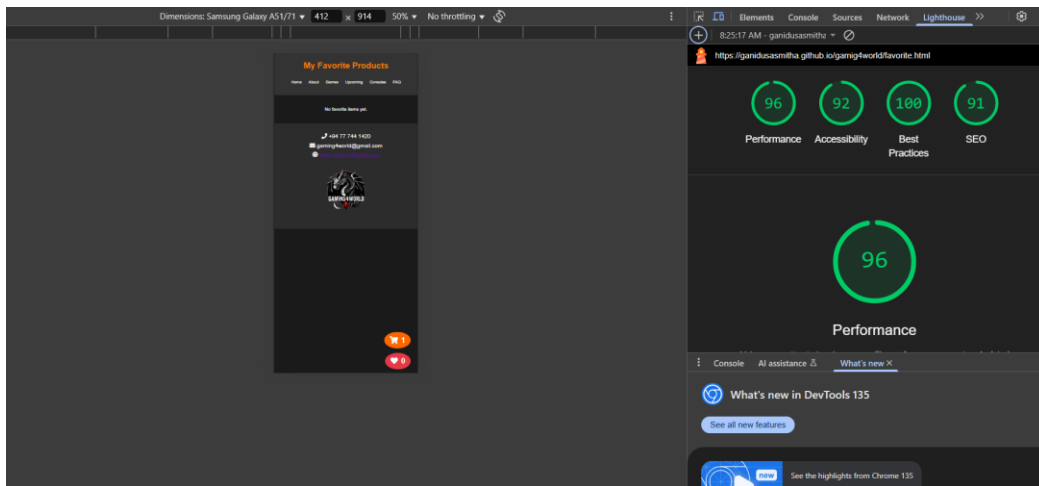


Figure 23

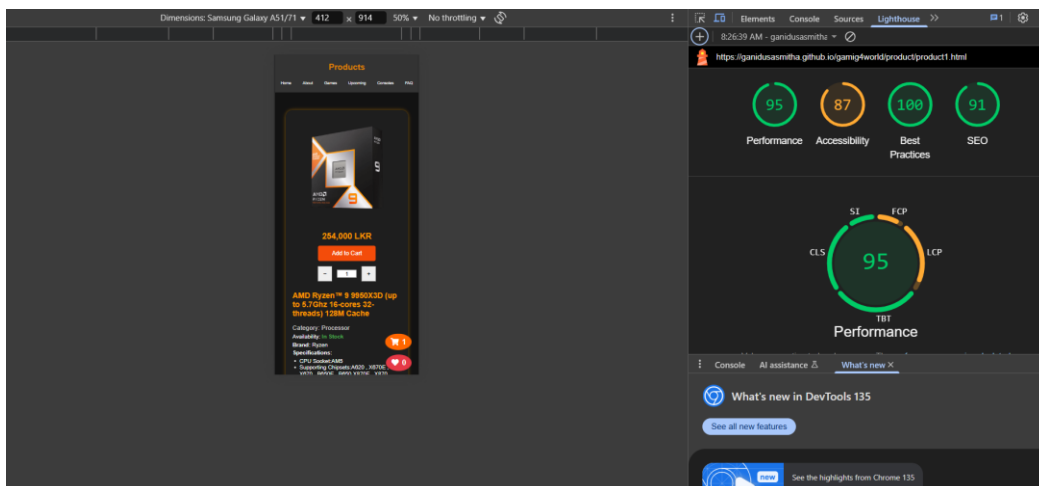


Figure 24

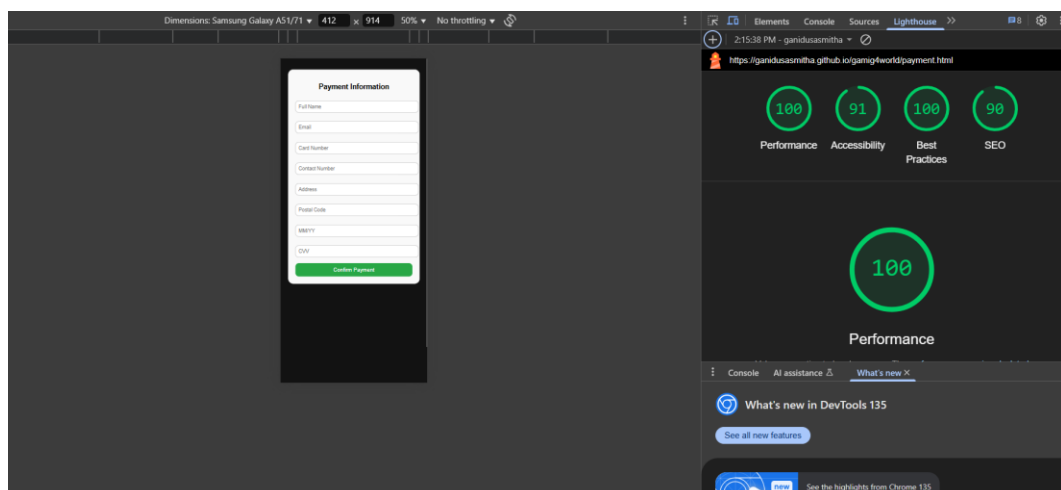


Figure 25

5.2 Mobile view tests

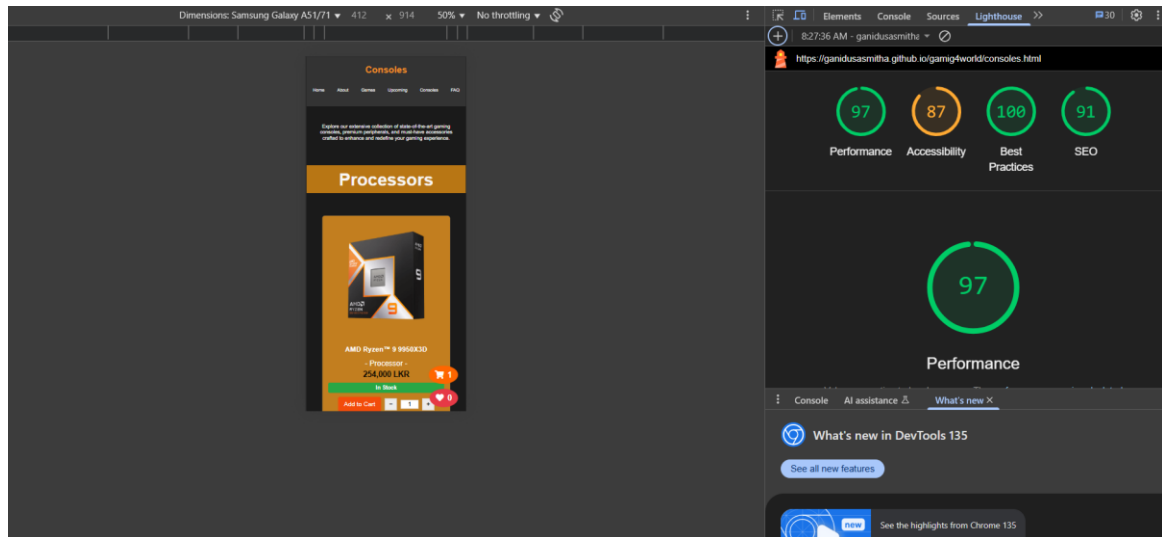


Figure 26

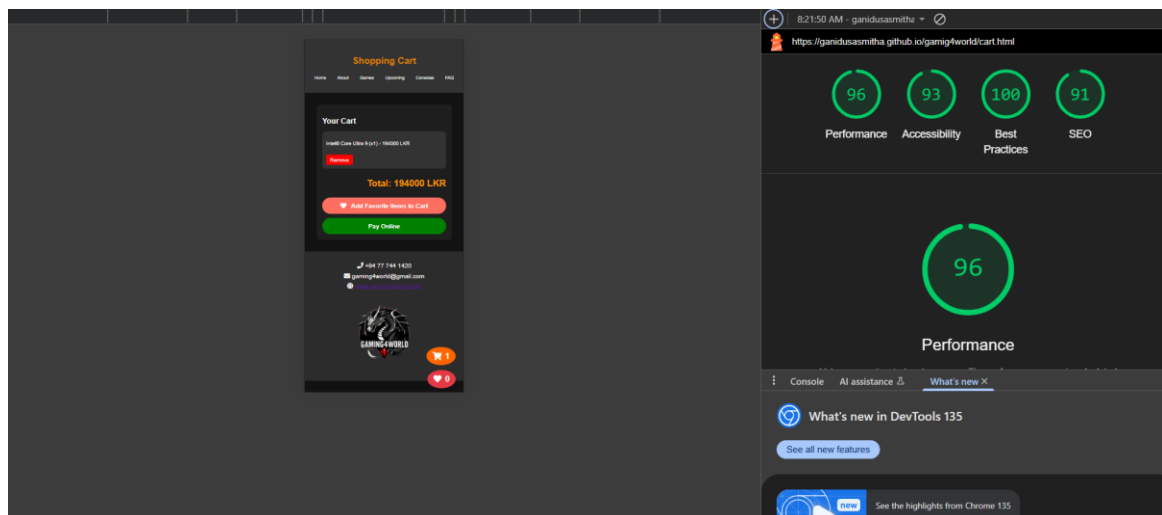


Figure 27

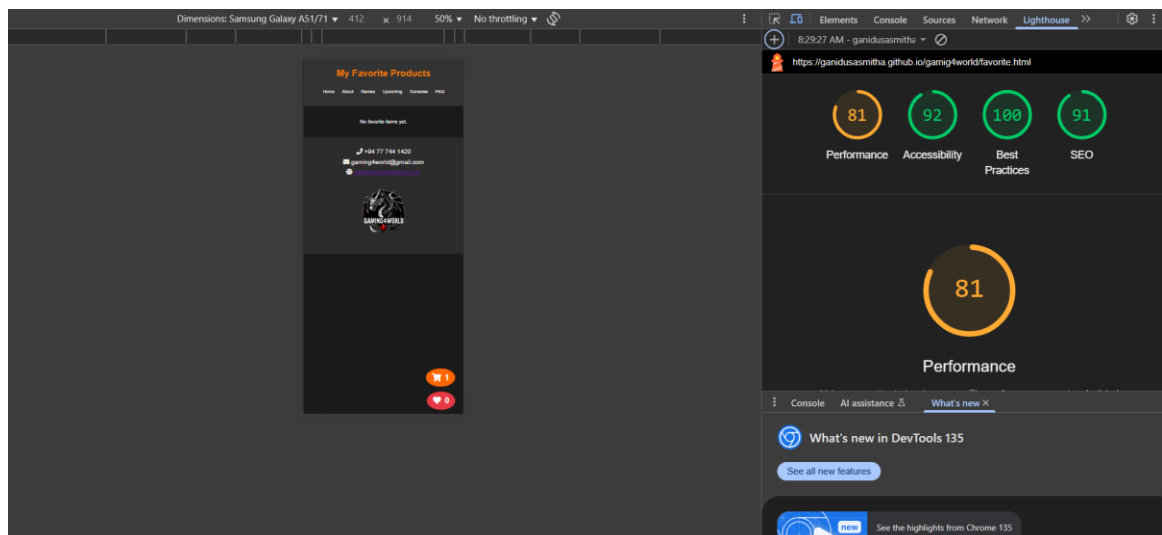


Figure 28

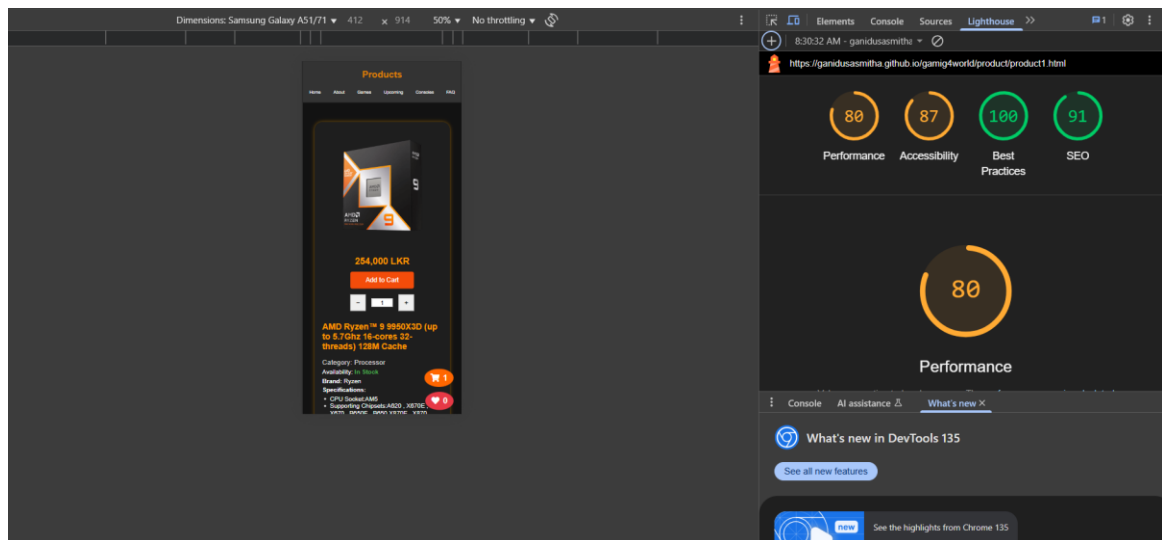


Figure 29

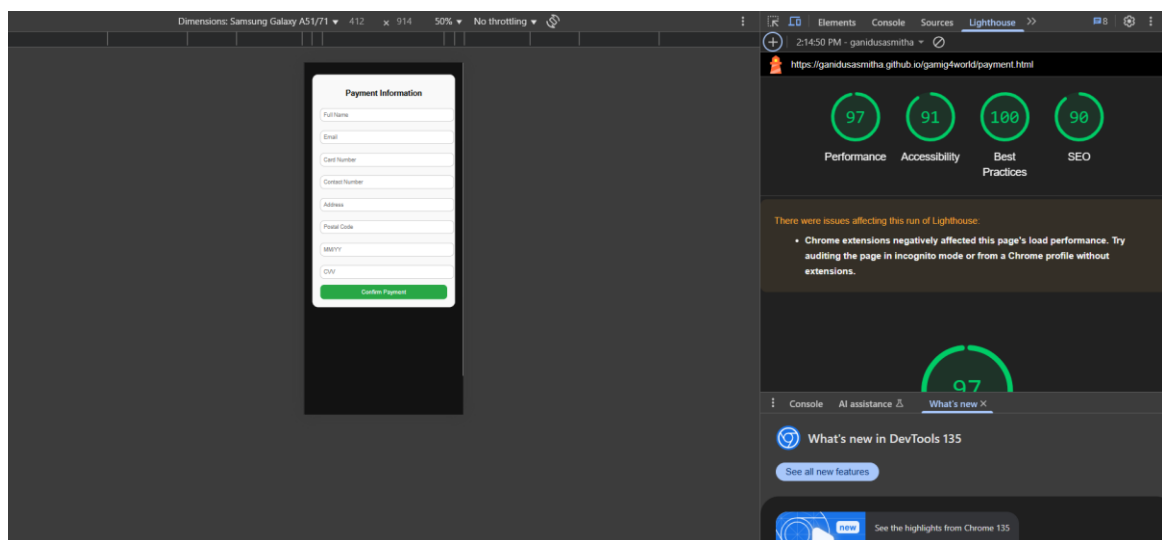


Figure 30

6. Responsiveness on different devices and browsers

6.1 Mobile view

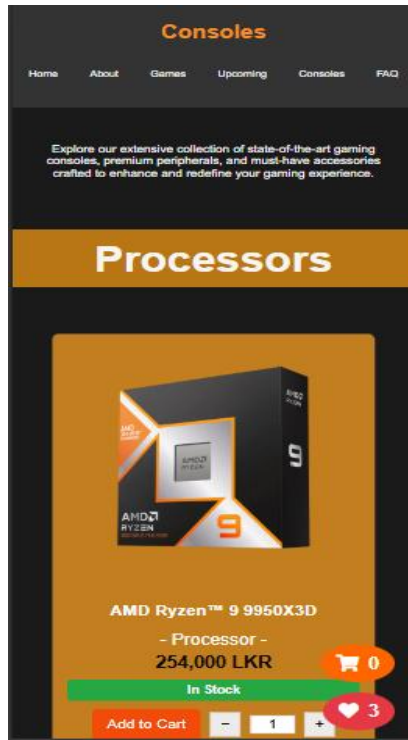


Figure 31

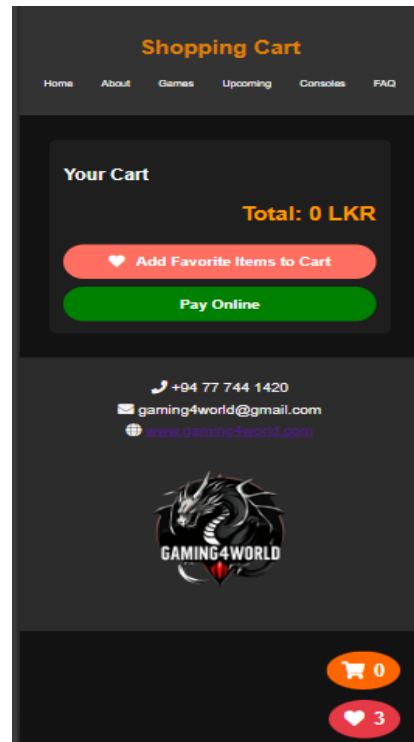


Figure 32



Figure 33

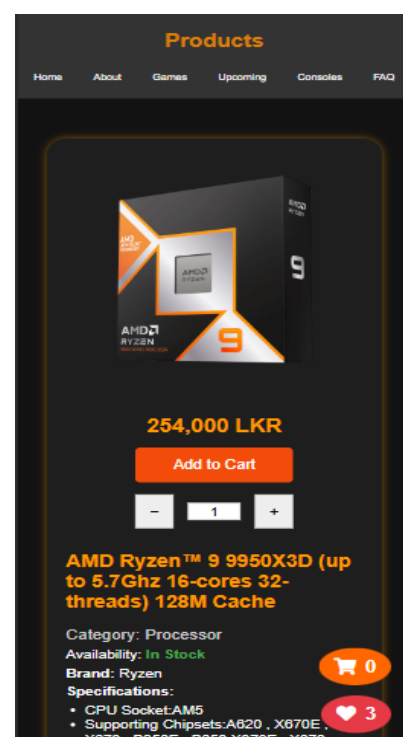


Figure 34

6.2 Tab view

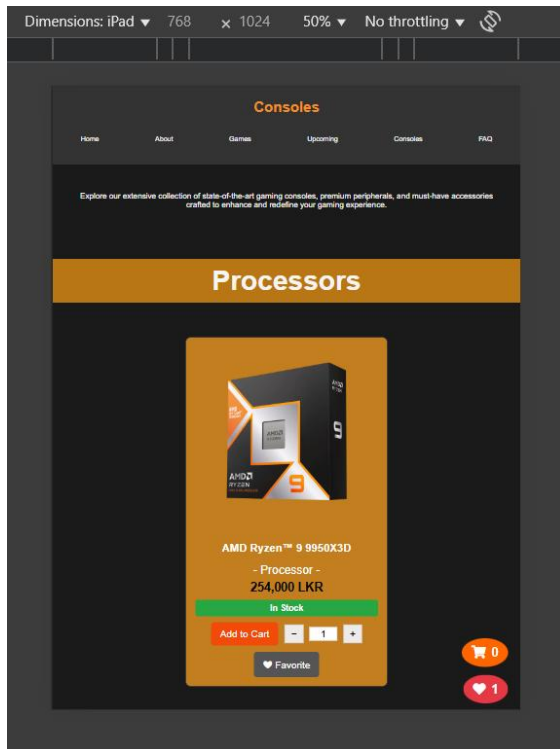


Figure 35

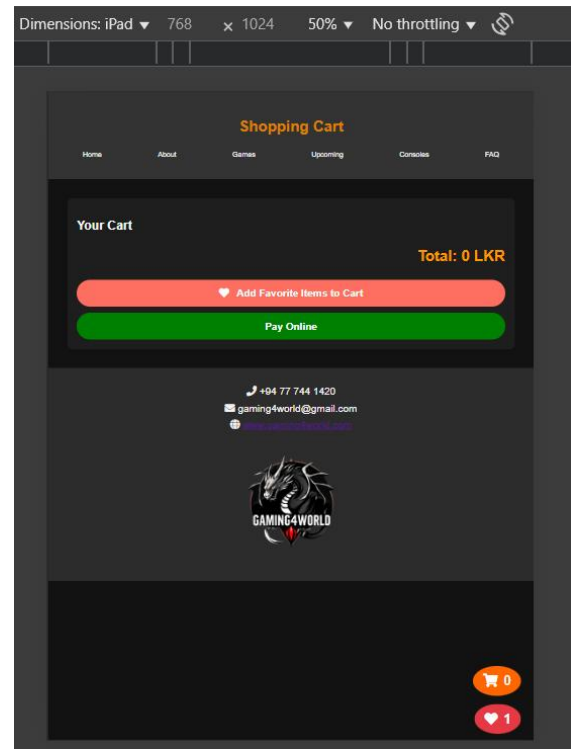


Figure 36

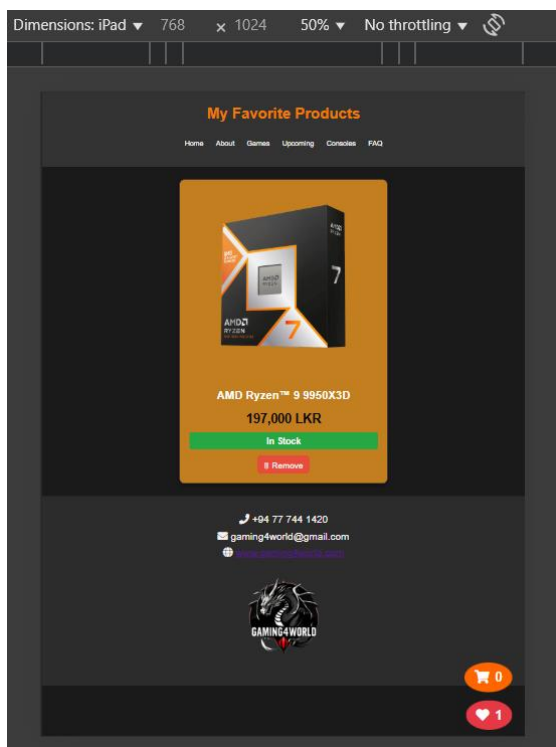


Figure 37

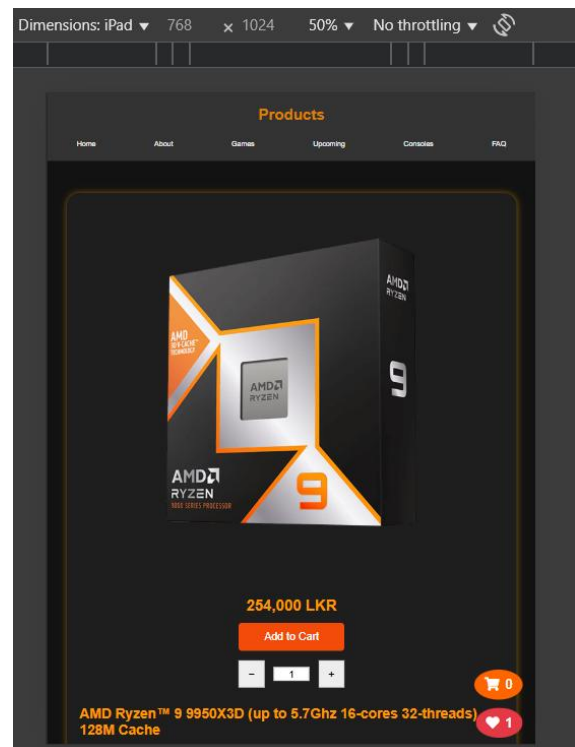


Figure 38

6.3 Edge browser view

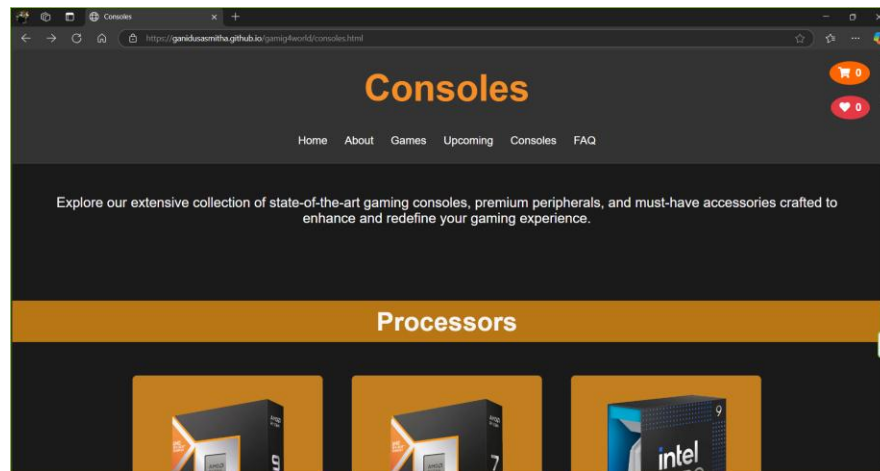


Figure 39

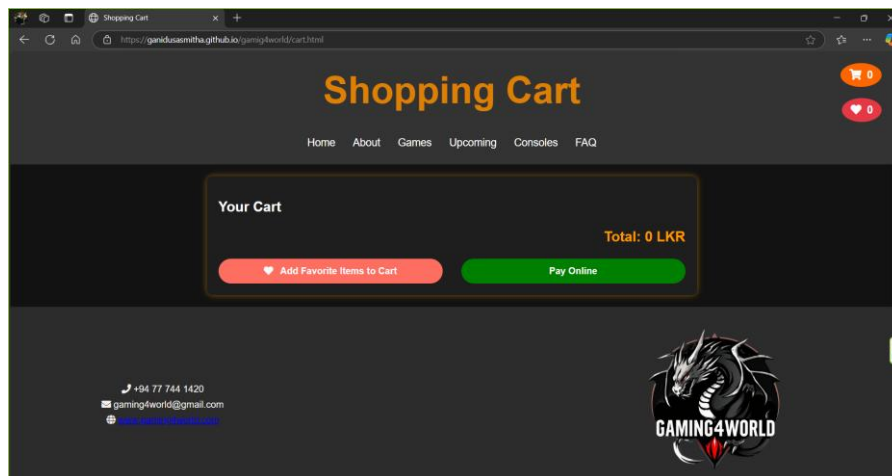


Figure 40

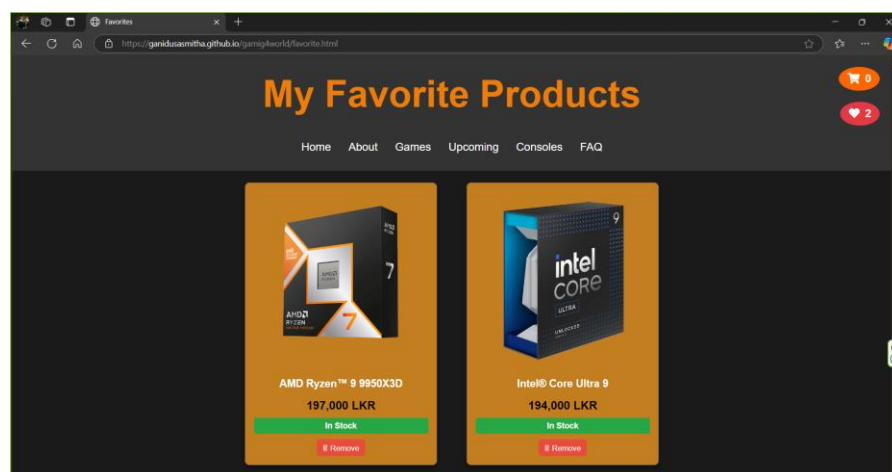


Figure 41

7. Functional tests for your product reservation application

Functionality testing was conducted on the Java product reservation application to verify that all the main user functions were working as required. These involved operations like adding products to the cart, favorite item management, and online checkout through payment methods. Testing involved making sure that all sections behaved as expected in response to user actions and right and wrong actions were handled appropriately by the application. There was some issue during testing, especially with handling forms and cart updating. These were addressed by analyzing and enhancing the Java programming and event handling. This leveled out the application and made it more user-friendly.

Test	Steps	Expected Result	Status
Add product to cart	Go to product page → Click “Add to Cart”	Product appears in cart with correct details	Pass
Increase quantity in cart	In cart → Click “+” in product	Quantity increases, total price updates	Pass
Decrease quantity in cart	In cart → Click “-” next to product	Quantity decreases (minimum 1), total price updates	Pass
Add product to favorites	On product → Click “Favorite”	Product appears in favorites page	Pass
Remove product from cart	On Cart page → Click “Remove”	Product is removed from the cart	Pass
Remove the product from Favorite	On favorites page → Click “Remove”	Product is removed from favorites	Pass
Proceed to checkout	In cart → Click “Checkout”	Navigated to payment page	Pass
Submit payment with valid details	Enter valid card info → Click “Pay”	Payment success message shows	Pass
Submit payment with invalid input	Leave fields empty or enter invalid card info → Click “Pay”	Error message appears, inputs invalid	Pass
Access empty favorites	Open favorites with no items	Message like “Favorite is empty” shows	Pass

8. Conclusion

Generally, this assignment was focused on the development and deployment of a web application for selling gaming products online with major functionalities such as product browsing, management of carts, favorite products, and payment through the internet. In the course of the project, major topics of web development were explored and applied, which included HTML and CSS validation of code quality, accessibility testing via WAVE to ensure accessible design, and measuring performance via Lighthouse testing.

Additionally, responsiveness was also tested on different devices and browsers to ascertain consistent user experience. Functional testing was performed to confirm fundamental features, bugs, and fixes needed were validated. Overall, this assignment improved practical knowledge of full-stack web development and highlighted the importance of testing and user-friendly design in the development of reliable, accessible, and user-friendly web applications.